

NICHOL FLOWERS

312.316.8282 | [Email](#) | [LinkedIn](#) | [Projects](#)

SUMMARY

Applied AI and Data Science professional with graduate training in Machine Learning, Generative AI, Data Engineering, and Statistics. Experience developing and optimizing machine learning models and Generative AI applications through experimentation and quantitative analysis. Brings a research-driven approach to investigating complex problems and translating findings into practical solutions.

EDUCATION

University of California - Berkeley

Master of Information and Data Science

Texas Southern University

Bachelor of Science, Chemistry

TECHNICAL SKILLS

Machine Learning, Deep Learning, Neural Networks, LLMs, Generative AI, Model Evaluation, Prompt Engineering, TensorFlow, Hugging Face, Computer Vision, Reinforcement Learning, Python, SQL, R, Data Engineering, Statistics, Neo4j, AWS, Docker, D3.js, Tableau

PROJECTS

[Eye2Voice: Gaze-to-Speech](#), UC Berkeley School of Information

- Developed a real-time gaze-to-speech system enabling individuals with severe physical disabilities to communicate using eye movements.
- Applied deep learning and computer vision techniques to translate gaze patterns into spoken output.
- Hal R. Varian MIDS Capstone Award winner, awarded for top data science capstone project

[Retrieval-Augmented Generation for Multiple Audiences](#), UC Berkeley School of Information

- Evaluated embeddings, chunking strategies, retrieval configurations, reranking approaches, and generation models.
- Developed an evaluation framework using LLM-as-a-Judge and retrieval-based metrics.
- Balanced evaluation rigor against token consumption and operational cost during large-scale testing.

[Second-Price Auction Strategy](#), UC Berkeley School of Information

- Developed a reinforcement learning-based bidding strategy for a simulated second-price advertising auction.
- Designed algorithms to infer user value from observed behavior and optimize bidding decisions.
- Recognized as Top Strategy among competing solutions

[Predicting California Wildfires](#), UC Berkeley School of Information

- Built an 8-class wildfire-occurrence classifier spanning 3-, 7-, and 14-day forecast horizons over 40+ years of California weather and fire data, beating a 12.5% random baseline.
- Engineered a geospatially enriched forecasting dataset (20,710 rows, 100+ features) by merging weather, fire-perimeter, and climate-division data with GeoPandas and Shapely, including lagged temporal features.
- Tuned XGBoost via randomized search under strict temporal validation, then identified a shared ~50% accuracy ceiling across model families that isolated data and feature quality, not model choice, as the binding constraint.

PROFESSIONAL EXPERIENCE

Cloud Engineer - Problem Management

2021 - current

Cisco Systems, Remote (San Jose)

- Investigate complex service incidents and recurring operational issues across large-scale collaboration platforms
- Analyze patterns within operational data to identify systemic causes of failures and reliability risks.
- Partner with engineering teams to drive long-term corrective actions and service improvements.

Global Problem Manager

2012 - 2017

HERE Technologies (formerly Navteq & Nokia), Chicago, IL

- Improved IT service availability by 35.8% by reducing the volume and impact of service disruptions through root cause analysis.
- Led the transition of Problem Management from manual reporting to real-time automated dashboards, empowering internal stakeholders with a self-service reporting platform.

Process Improvement Lead

2010 - 2012

- Pioneered Navteq's first product-specific strategy for collecting and updating map data for GPS products, contributing to an \$8.5 million cost savings across the product portfolio.
- Optimized processes for discovering global map changes faster reducing manual data collection time by 50%.

Volunteer Research Associate

2017 - 2018

University of Illinois - Chicago

Chicago, IL

- Improved data quality and consistency for federally funded nutrition and public health research studies.